

NanoGT Match Report: Nephropathic cystinosis

Disease: Nephropathic cystinosis (ORPHA:213)

Primary gene: CTNS

Gene CDS: 1104 bp

Inheritance: Autosomal recessive

Target tissues scored: kidney, multisystem

Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: LYSOSOMAL MEMBRANE PROTEIN (CTNS): the disease gene encodes a lysosomal membrane channel or transporter, NOT a soluble secretable enzyme. The HSC-lentiviral cross-correction strategy of precedent programs such as Libmeldy (ARSA) and Skysona (ABCD1) relies on enzyme secretion from microglia and M6P receptor-mediated uptake by neurons — a mechanism that is physically impossible for a membrane-anchored protein. Direct in vivo AAV delivery to individual target neurons is required. Precedent scores from HSC/LV programs compare delivery platform only, not therapeutic mechanism.; Vector does not naturally cover all annotated disease tissues: kidney, multisystem; No direct tissue overlap; treat this as weak precedent unless route or modality is changed.

Disease Mechanism Evidence

Molecular mechanism: loss of function

Mechanistic detail: Cystinosis lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs

Gene-addition compatibility: conditional

Preferred modality class: kidney or systemic gene addition

Evidence level/status: direct / needs_user_fact_check

Evidence summary: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues

Evidence source: [OMIM CTNS 219800](#)

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and integrating ex vivo HSC vector precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - No catalog program directly targets the annotated disease tissue(s); ranking is extrapolating from indirect precedents.
 - No catalog vector naturally covers the annotated disease tissue(s); consider non-catalog vectors or alternative modalities.
 - Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.
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Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Libmeldy	LV	6.6/10	Medium	approved
2	AVR-RD-01	LV	6.2/10	Medium	phase1/2
3	ST-920	AAV2/6	6.0/10	Medium	phase1/2
4	Skysona	LV	6.0/10	Medium	approved
5	ABO-101	AAV9	5.9/10	Medium	phase1/2

Match #1: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.6 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.50	1.0	Immunological protection of target tissue
Promoter availability	0.50	1.0	Validated tissue-specific promoters exist
Route of administration	0.50	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.57	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1104bp / cargo 8000bp (14% utilized)
- No tissue overlap (disease: ['kidney', 'multisystem'], vector: ['hematopoietic'])

- Lysosomal pathway shared, but disease gene encodes a lysosomal MEMBRANE protein (channel/transporter), NOT a soluble enzyme. The HSC-mediated cross-correction mechanism of the matched precedent is not applicable. Direct per-cell vector delivery to every target neuron is required.
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: leukodystrophy
- Disease mechanism: loss of function — Cystinosis lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues
- Mechanism source: OMIM CTNS 219800 (<https://omim.org/entry/219800>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Lysosomal MEMBRANE protein (channel/transporter) — no cross-correction possible. The protein is anchored in the lysosomal membrane and cannot be released as soluble cargo. Every target cell must individually receive the vector. HSC-based lentiviral programs (Libmeldy) depend on secreted enzyme cross-correction and are not applicable as direct strategies for this protein class.
- Immune privilege: moderate-low privilege; renal immune surveillance is significant
- Promoter availability: Limited clinical-grade promoters validated for multisystem
- Route of administration: Delivery route to multisystem not well established in clinical programs
- ORGANELLE TARGETING: COMPATIBLE — CTNS standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- LYSOSOMAL MEMBRANE PROTEIN (CTNS): the disease gene encodes a lysosomal membrane channel or transporter, NOT a soluble secretable enzyme. The HSC-lentiviral cross-correction strategy of precedent programs such as Libmeldy (ARSA) and Skysona (ABCD1) relies on enzyme secretion from microglia and M6P receptor-mediated uptake by neurons — a mechanism that is physically impossible for a membrane-anchored protein. Direct in vivo AAV delivery to individual target neurons is required. Precedent scores from HSC/LV programs compare delivery platform only, not therapeutic mechanism.
- Vector does not naturally cover all annotated disease tissues: kidney, multisystem
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable

Match #2: AVR-RD-01

Precedent disease: Fabry disease

Vector: LV

Tissue target: hematopoietic

Composite score: 6.2 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.50	1.0	Immunological protection of target tissue
Promoter availability	0.50	1.0	Validated tissue-specific promoters exist
Route of administration	0.50	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.19	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1104bp / cargo 8000bp (14% utilized)
- No tissue overlap (disease: ['kidney', 'multisystem'], vector: ['hematopoietic'])
- Lysosomal pathway shared, but disease gene encodes a lysosomal MEMBRANE protein (channel/transporter), NOT a soluble enzyme. The HSC-mediated cross-correction mechanism of the matched precedent is not applicable. Direct per-cell vector delivery to every target neuron is required.
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal storage
- Disease mechanism: loss of function — Cystinosis lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues
- Mechanism source: OMIM CTNS 219800 (<https://omim.org/entry/219800>)
- Approval status: phase1/2
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Lysosomal MEMBRANE protein (channel/transporter) — no cross-correction possible. The protein is anchored in the lysosomal membrane and cannot be released as soluble cargo. Every target cell must individually receive the vector. HSC-based lentiviral programs (Libmeldy)

depend on secreted enzyme cross-correction and are not applicable as direct strategies for this protein class.

- Immune privilege: moderate-low privilege; renal immune surveillance is significant
- Promoter availability: Limited clinical-grade promoters validated for multisystem
- Route of administration: Delivery route to multisystem not well established in clinical programs
- ORGANELLE TARGETING: COMPATIBLE — CTNS standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- LYSOSOMAL MEMBRANE PROTEIN (CTNS): the disease gene encodes a lysosomal membrane channel or transporter, NOT a soluble secretable enzyme. The HSC-lentiviral cross-correction strategy of precedent programs such as Libmeldy (ARSA) and Skysona (ABCD1) relies on enzyme secretion from microglia and M6P receptor-mediated uptake by neurons — a mechanism that is physically impossible for a membrane-anchored protein. Direct in vivo AAV delivery to individual target neurons is required. Precedent scores from HSC/LV programs compare delivery platform only, not therapeutic mechanism.
- Vector does not naturally cover all annotated disease tissues: kidney, multisystem
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable

Match #3: ST-920

Precedent disease: Fabry disease

Vector: AAV2/6

Tissue target: liver

Composite score: 6.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	1.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.50	1.0	Immunological protection of target tissue
Promoter availability	0.50	1.0	Validated tissue-specific promoters exist
Route of administration	0.50	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.95	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1104bp / cargo 4700bp (23% utilized)
- No tissue overlap (disease: ['kidney', 'multisystem'], vector: ['liver', 'muscle'])
- Lysosomal pathway shared, but disease gene encodes a lysosomal MEMBRANE protein (channel/transporter), NOT a soluble enzyme. The HSC-mediated cross-correction mechanism of the matched precedent is not applicable. Direct per-cell vector delivery to every target neuron is required.
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal storage
- Disease mechanism: loss of function — Cystinosis lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues
- Mechanism source: OMIM CTNS 219800 (<https://omim.org/entry/219800>)
- Approval status: phase1/2
- Vector immunogenicity (AAV2/6): moderate (~17%) — significant proportion may require pre-screening or exclusion
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Lysosomal MEMBRANE protein (channel/transporter) — no cross-correction possible. The protein is anchored in the lysosomal membrane and cannot be released as soluble cargo. Every target cell must individually receive the vector. HSC-based lentiviral programs (Libmeldy) depend on secreted enzyme cross-correction and are not applicable as direct strategies for this protein class.
- Immune privilege: moderate-low privilege; renal immune surveillance is significant
- Promoter availability: Limited clinical-grade promoters validated for multisystem
- Route of administration: Delivery route to multisystem not well established in clinical programs
- ORGANELLE TARGETING: COMPATIBLE — CTNS standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- LYSOSOMAL MEMBRANE PROTEIN (CTNS): the disease gene encodes a lysosomal membrane channel or transporter, NOT a soluble secretable enzyme. The HSC-lentiviral cross-correction strategy of precedent programs such as Libmeldy (ARSA) and Skysona (ABCD1) relies on enzyme secretion from microglia and M6P receptor-mediated uptake by neurons — a mechanism

that is physically impossible for a membrane-anchored protein. Direct in vivo AAV delivery to individual target neurons is required. Precedent scores from HSC/LV programs compare delivery platform only, not therapeutic mechanism.

- Vector does not naturally cover all annotated disease tissues: kidney, multisystem
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.50	1.0	Immunological protection of target tissue
Promoter availability	0.50	1.0	Validated tissue-specific promoters exist
Route of administration	0.50	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.95	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1104bp / cargo 8000bp (14% utilized)
- No tissue overlap (disease: ['kidney', 'multisystem'], vector: ['hematopoietic'])
- Both membrane proteins
- LOF inheritance — compatible for gene replacement
- Different pathway (lysosomal_storage vs peroxisomal)
- Disease mechanism: loss of function — Cystinosis lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues
- Mechanism source: OMIM CTNS 219800 (<https://omim.org/entry/219800>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Lysosomal MEMBRANE protein (channel/transporter) — no cross-correction possible. The protein is anchored in the lysosomal membrane and cannot be released as soluble cargo. Every target cell must individually receive the vector. HSC-based lentiviral programs (Libmeldy) depend on secreted enzyme cross-correction and are not applicable as direct strategies for this protein class.
- Immune privilege: moderate-low privilege; renal immune surveillance is significant
- Promoter availability: Limited clinical-grade promoters validated for multisystem
- Route of administration: Delivery route to multisystem not well established in clinical programs
- ORGANELLE TARGETING: COMPATIBLE — CTNS standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- LYSOSOMAL MEMBRANE PROTEIN (CTNS): the disease gene encodes a lysosomal membrane channel or transporter, NOT a soluble secretable enzyme. The HSC-lentiviral cross-correction strategy of precedent programs such as Libmeldy (ARSA) and Skysona (ABCD1) relies on enzyme secretion from microglia and M6P receptor-mediated uptake by neurons — a mechanism that is physically impossible for a membrane-anchored protein. Direct in vivo AAV delivery to individual target neurons is required. Precedent scores from HSC/LV programs compare delivery platform only, not therapeutic mechanism.
- Vector does not naturally cover all annotated disease tissues: kidney, multisystem
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable

Match #5: ABO-101

Precedent disease: Mucopolysaccharidosis type IIIB

Vector: AAV9

Tissue target: CNS

Composite score: 5.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.50	1.0	Immunological protection of target tissue
Promoter availability	0.50	1.0	Validated tissue-specific promoters exist
Route of administration	0.50	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.86	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1104bp / cargo 4700bp (23% utilized)
- No tissue overlap (disease: ['kidney', 'multisystem'], vector: ['CNS', 'muscle', 'liver', 'heart'])
- Lysosomal pathway shared, but disease gene encodes a lysosomal MEMBRANE protein (channel/transporter), NOT a soluble enzyme. The HSC-mediated cross-correction mechanism of the matched precedent is not applicable. Direct per-cell vector delivery to every target neuron is required.
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: lysosomal storage
- Disease mechanism: loss of function — Cystinosis lysosomal cystine transporter deficiency — membrane protein; per-cell delivery required across kidney and multiple organs
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: CTNS encodes a lysosomal membrane transporter; cross-correction via M6P uptake is not applicable — per-cell delivery needed in target tissues
- Mechanism source: OMIM CTNS 219800 (<https://omim.org/entry/219800>)
- Approval status: phase1/2
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Lysosomal MEMBRANE protein (channel/transporter) — no cross-correction possible. The protein is anchored in the lysosomal membrane and cannot be released as soluble cargo. Every

target cell must individually receive the vector. HSC-based lentiviral programs (Libmeldy) depend on secreted enzyme cross-correction and are not applicable as direct strategies for this protein class.

- Immune privilege: moderate-low privilege; renal immune surveillance is significant
- Promoter availability: Limited clinical-grade promoters validated for multisystem
- Route of administration: Delivery route to multisystem not well established in clinical programs
- ORGANELLE TARGETING: COMPATIBLE — CTNS standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- LYSOSOMAL MEMBRANE PROTEIN (CTNS): the disease gene encodes a lysosomal membrane channel or transporter, NOT a soluble secretable enzyme. The HSC-lentiviral cross-correction strategy of precedent programs such as Libmeldy (ARSA) and Skysona (ABCD1) relies on enzyme secretion from microglia and M6P receptor-mediated uptake by neurons — a mechanism that is physically impossible for a membrane-anchored protein. Direct in vivo AAV delivery to individual target neurons is required. Precedent scores from HSC/LV programs compare delivery platform only, not therapeutic mechanism.
- Vector does not naturally cover all annotated disease tissues: kidney, multisystem
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone